YAU, Chung Yiu

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RESEARCH EXPERIENCE

- (TMLR) (NeurIPS 2026) Decentralized optimization. Optimization on large-scale datasets for parallel computation and minimal communication.
- (NeurIPS 2026) Accelerated optimization algorithms for deep learning.
- (ICML 2025) Large language model quantization for memory efficiency and inference speedup.
- (ICML 2024) Optimization for contrastive learning and negative sampling.

EDUCATION & ACADEMIA

Postdoctoral Associate **Nov 2025 - Later**
@ Department of Electrical and Computer Engineering, University of Minnesota.
Advisor: Prof. Mingyi Hong.

- Research focus: optimization algorithms for large language model pre-training.

Ph.D. Systems Engineering & Engineering Management **2021 - 2025**
@ The Chinese University of Hong Kong.
Supervisor: Prof. Hoi-To Wai.

- Research focus: convergence analysis of new decentralized optimization algorithms with communication compression.

B.Sc. Computer Science **2017 - 2021**
@ The Chinese University of Hong Kong.

- First Class Honour, ELITE Stream.

WORK EXPERIENCE

Applied Scientist Intern **June - Sep 2024**
@ Amazon Web Services, Santa Clara, California.

- Research on large language model quantization.
- Proposed an adaptive rotation method with paper presented at ICML 2025. Our algorithm fine-tunes a quantized LLM by quantization-aware training on Llama, Pythia and Qwen models.

Applied Scientist Intern **June - August 2023**
@ Amazon Web Services, Shanghai.

- Study the large batch dependency in contrastive learning.
- Proposed a small batch sampling algorithm with paper presented at ICML 2024. Our algorithm improves negative pair sampling and achieves exact convergence.

ACADEMIC ACTIVITIES

Invited Talk

Communication Efficient Decentralized Optimization **Sep 2024**
@ Department of Electrical and Computer Engineering, University of Minnesota.

LLM Quantization-Aware Training **July 2025**
@ INFORMS International 2025, Singapore.

Teaching Assistant **2021 - 2025**
@ Department of Systems Engineering and Engineering Management, The Chinese University of Hong Kong.

- ◇ ENGG2440 Discrete Mathematics for Engineers.
- ◇ FTEC2101 Optimization Methods.

Reviewer
@ conferences and journals:

- ◇ NeurIPS (2025 Top Reviewer, 2026), ICML (2024, 2025, 2026), ICLR (2025, 2026), TMLR.
- ◇ IEEE TSP, ICASSP, TAC, L-CSS.

Area Chair @ ICLR 2027.

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RESEARCH PUBLICATION

[1] Zijian Zhang, Rizhen Hu, Athanasios Glentis, Dawei Li, Chung-Yiu Yau, Hongzhou Lin, and Mingyi Hong. Is one layer enough? training a single transformer layer can match full-parameter rl training. *arXiv preprint arXiv:2607.01232*, 2026.

[2] Chung-Yiu Yau, Dawei Li, Athanasios Glentis, Valentyn Boreiko, Hoi-To Wai, and Mingyi Hong. Ema-nesterov: Stabilizing nesterov's lookahead for accelerated deep learning optimization. *Advances in Neural Information Processing Systems*, 2026.

[3] Athanasios Glentis, Dawei Li, Chung-Yiu Yau, and Mingyi Hong. What stops sgd on llm pre-training: The need for large learning rate and how to achieve it. *Advances in Neural Information Processing Systems*, 2026.

[4] Chung-Yiu Yau, Haoming Liu, and Hoi-To Wai. Stochastic approximation approach for decentralized optimization on time varying random networks. *Advances in Neural Information Processing Systems*, 2026.

[5] Haoming Liu, Chung-Yiu Yau, and Hoi-To Wai. Decentralized stochastic optimization over unreliable networks via two-timescales updates. *IEEE Transactions on Signal Processing*, 74:121–136, 2025.

[6] Quan Wei*, Chung-Yiu Yau*, Hoi-To Wai, Dongyeop Kang, Youngsuk Park, Mingyi Hong, et al. Roste: An efficient quantization-aware supervised fine-tuning approach for large language models. *Proceedings of the 42st International Conference on Machine Learning*, 2025. *Equal contribution.

[7] Haoming Liu, Chung-Yiu Yau, and Hoi-To Wai. A two-timescale primal-dual algorithm for decentralized optimization with compression. In *2025 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2025.

[8] Chung-Yiu Yau, Hoi-To Wai, Parameswaran Raman, Soumajyoti Sarkar, and Mingyi Hong. EMC^2: Efficient MCMC negative sampling for contrastive learning with global convergence. In *Proceedings of the 41st International Conference on Machine Learning*, pages 56966–56981. PMLR, 2024.

[9] Chung-Yiu Yau and Hoi-To Wai. Fully stochastic distributed convex optimization on time-varying graph with compression. In *2023 62nd IEEE Conference on Decision and Control (CDC)*, pages 145–150. IEEE, 2023.

[10] Xiaolu Wang, Chung-Yiu Yau, and Hoi-To Wai. Network effects in performative prediction games. In *International Conference on Machine Learning*, pages 36514–36540. PMLR, 2023.

[11] Chung-Yiu Yau and Hoi-To Wai. Docom: Compressed decentralized optimization with near-optimal sample complexity. *Transactions on Machine Learning Research*, 2023.

[12] Bingqing Song, Ioannis Tsaknakis, Chung-Yiu Yau, Hoi-To Wai, and Mingyi Hong. Distributed Optimization for Overparameterized Problems: Achieving Optimal Dimension Independent Communication Complexity. *Advances in Neural Information Processing Systems*, 2022.

[13] Qiang Li, Chung-Yiu Yau, and Hoi-To Wai. Multi-agent Performative Prediction with Greedy Deployment and Consensus Seeking Agents. *Advances in Neural Information Processing Systems*, 2022.

[14] Chung-Yiu Yau, Haoli Bai, Irwin King, and Michael R Lyu. DAP-BERT: Differentiable Architecture Pruning of BERT. In *International Conference on Neural Information Processing*, pages 367–378. Springer, 2021.